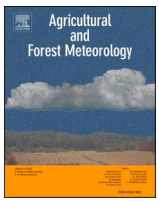
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Overcoming mechanistic limitations of process-based phenological models: A data clustering method for large-scale applications

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A R T I C L E I N F O

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Model comparison

A B S T R A C T

Process-based phenological models use thermal requirement (TR) defined by planting date, temperature and photoperiod to predict crop developmental stages. The TR value for a specific developmental stage for a given variety is often presumed to be constant regardless of environmental conditions. We calibrated and compared 12 phenological models using 27-year of observation data of one unique rice (*Oryza sativa* L.) variety ('Shanyou63') from 46 sites in southern China. Our findings indicated that shifts in environmental conditions significantly affected TR values, e.g., standard deviations of TR at physiological maturation ranged from 83 to 167 °C d. Clustering sites together minimized environmental heterogeneity, and thus minimized the differences in TR for different phenological stages. When the increased from 1 to 24, simulation errors for the 12 models showed a significant decrease across all developmental stages, from 1.8 to 1.4 days for tillering, from 5.3 to 3.7 days for jointing, from 5.6 to 3.9 days for booting, from 4.7 to 3.3 days for heading, and from 6.6 to 3.7 days for physiological maturation. Furthermore, our findings indicate that models featuring a three-segment piecewise linear temperature response function provide a more precise prediction. In contrast, models incorporating a Beta temperature response function have not performed well. This difference is attributed to different mechanisms used to describe the response of crop development rate to temperature, particularly at non-optimum temperatures. The impact of the photoperiod response function on prediction accuracy became significant with the expansion of scale. Our results demonstrate that clustering method effectively compensates for the lack of crop adaptation processes in common phenological models, leading to significantly improved phenological prediction accuracy in regions with environmental diversity.

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1. Introduction

Process-based crop models serve as valuable tools for optimizing agricultural management practices, aiming to enhance the efficient utilization of land, water, and other natural resources for food production (Food and Agriculture Organization of the United Nations, 2017). A phenological model, as a timer of crop growth, is an essential component of crop models that regulates the occurrences of different phenological events and impacts the mass and energy cycles of cropping systems (Ahmad et al., 2012; Steduto et al., 2012; Sulṭānī and Sinclair, 2012; Zhao et al., 2013). Therefore, the accuracy of phenological models has direct impacts on crop model performance during the optimization of management practices (Wu et al., 2017; Zhang and Tao, 2013; Zhang et al., 2016, 2008) and on climate change impact assessment and adaptation investigations (Wakatsuki et al., 2023; Wang et al., 2015; Wassmann et al., 2009; Wu et al., 2019a; Zhang and Tao, 2013). Furthermore, phenological models play a crucial role in scheduling timely planting and harvesting of crops, determining the appropriate timing and application rates of fertilizers and pesticides, and adjusting crop varieties to adapt to climate change (Porker et al., 2020; Seyoum et al., 2018; Zhao et al., 2023, 2022).

Crop phenological development is predominantly governed by temperature and photoperiod, and to a lesser extent by drought and nutrition/fertility (Mcmaster, 1997; Pan and Su, 2011; Zhao et al., 2013). The effects of drought and nutrient stresses usually impact phenological development only in some situations (Lilley and Fukai, 1994; Passioura, 2002), so they are often ignored in crop models. Regarding temperature, the concept of thermal requirement (TR), i.e., the thermal time required to reach a developmental stage, has been widely used to quantify phenological development (Atkinson and R. Porter, 1996; McMaster, 1997; Ritchie and Nesmith, 1991). This concept assumes certain response curves between temperature and crop development rate. These response curves are defined by cardinal temperatures and can be modified by a crop's sensitivity to photoperiod and vernalization requirement, namely photo-thermal and photo-vernal-thermal sensitivity (Basso et al., 2016; Boogaard et al., 2014; Bouman et al., 2001; Keating et al., 2003; Wang and Engel, 1998). Cardinal temperatures and developmental stage-wise TR are used as parameters in these models. For a certain crop variety, the TR parameters are generally assumed to be constant and linked to genetic characteristics, and therefore, do not change across environmental conditions. This concept functions well when the model is calibrated and used at a single site or in relatively homogeneous regions, but it may lead to large errors over large areas with significant environmental variability, such as occurs in simulation studies at continental or global scales (van Oort et al., 2011; Wu et al., 2019b).

Despite the fact that site-specific variations of TRs are frequently reported for various crops from observations at different sites (Fadón et al., 2020; Gannouni et al., 2017; Major et al., 1983; Tataryn, 1974; Wu et al., 2019b; Zhang et al., 2008), the concept of cultivar-specific parameterization of TR has not been evolved during the recent past. Although efforts have been made to limit environmental variability in the parameterization of crop models (Batchelor et al., 2002; Luo and Schuur, 2020; Zhao et al., 2016), these efforts have been based on the assumption that different cultivars will respond differently in different environmental zones (Iizumi et al., 2009; Li et al., 2015; Zhao et al., 2015b). One reason for this lack of progress in determining cultivar-specific parameterization of TR is that the uncertainty from a selected model cannot be excluded from previous studies (Campoy et al., 2012; Fadón et al., 2023; Raffaella Viti et al., 2010) when only one model was used in the calculation of TR in these studies. Secondly, only a limited number of observations have been used in these studies and the uncertainty from data cannot be ignored.

Therefore, we used 27 years of rice (*Oryza sativa* L.) phenology observations of one variety (i.e. Shanyou63) from 46 sites in southern China and 12 phenology models in the current study to determine: (1) the variability of TR across site-years and models; (2) the effects on model prediction errors of reducing environmental variability by a clustering method; (3) the applicability, limitations, and sources of uncertainty of these phenological models; and (4) a parameter optimization method to overcome the mechanistic limitations of phenological models for large-scale applications.

2. Materials and methods

2.1. Study area and data source

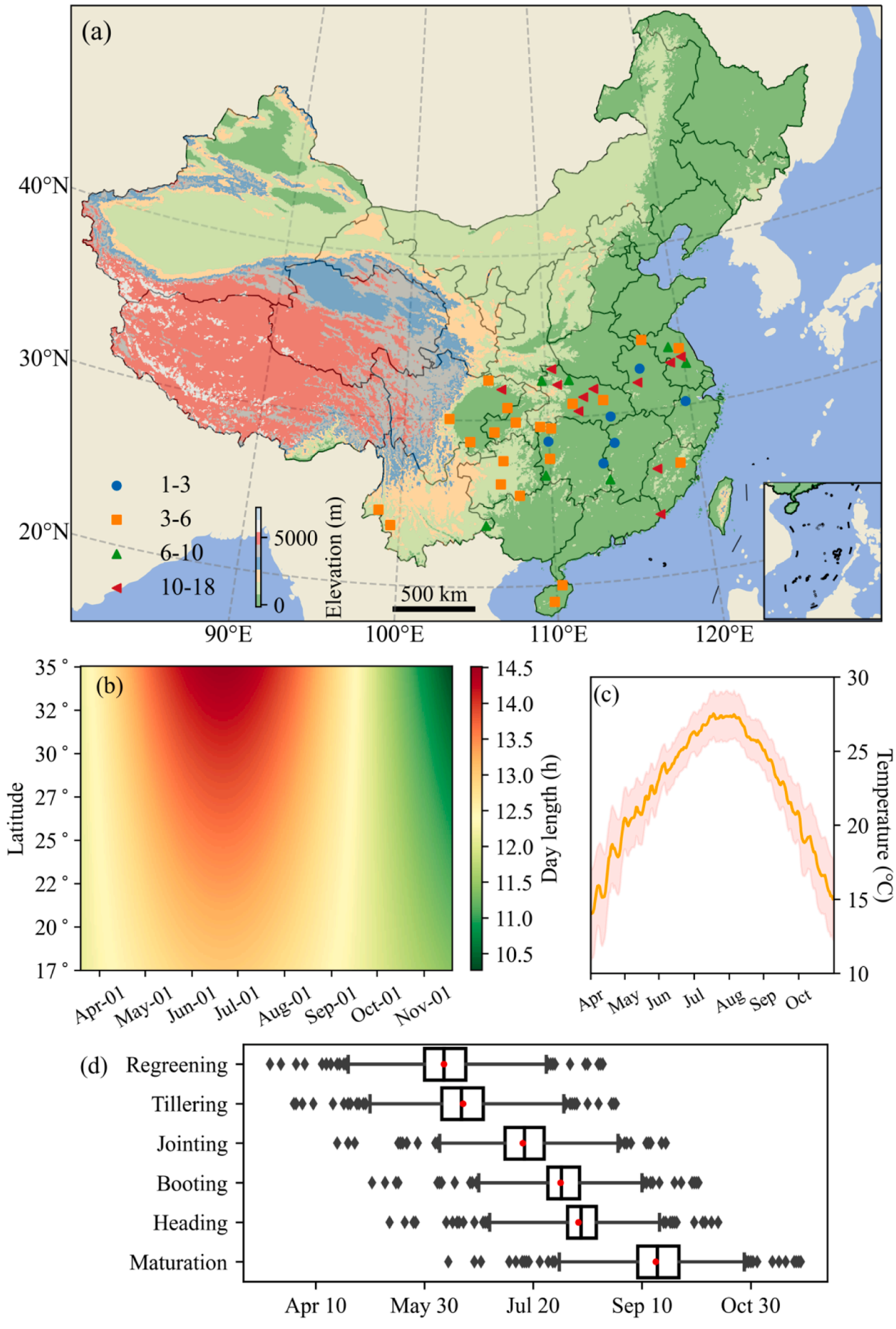
The study area is located in the rice cropping region of southern China. The area encompasses a large latitude span and fluctuating terrain (Fig. 1a), and accounts for 90 % of China's planting area and 88 % of China's total rice production (Yao and Zhang, 2007). The study area is dominated by a subtropical warm and humid monsoon climate, with diverse landscapes including sub-tropical plateaus, hilly areas, the southwest plateau, and plain areas in the middle and lower reaches of the Yangtze River. The elevation ranges from 3 m to 1105 m, and the latitude and longitude span from 19°N to 35°N and 98°E to 120°E, respectively, signifying diverse climates across this region. Across all sites, the average daylength ranges from 10.2 h to 14.5 h (Fig. 1b), and the daily mean temperature during the growing season varies from 13 °C to 28 °C (Fig. 1c).

Phenological observations from 1984 to 2010 of indica rice variety 'Shanyou63' ('SY63') were obtained from 46 agrometeorological stations (http://www.cma.gov.cn/). Key phenological dates were recorded: transplanting, regreening (Reg), tillering (Til), jointing (Joi), boot (Bot), heading (Hea), and physiological maturation (Mat). These dates were recorded when 50 % of the plants in the experimental field showed the specific designated phenological trait, according to the Argo-Meteorological Observation Standard (China Meteorological Administration, 1993). Only stages at and after regreening (Reg, Til, Joi, Bot, Hea, and Mat) were selected to eliminate the effect of transplanting shock. We then divided the duration between the phenological dates into five periods: Reg–Til, Til–Joi, Joi–Bot, Bot–Hea, and Hea–Mat. The meteorological dataset from 1984 to 2010 was obtained from the China Meteorological Data Sharing Service Website (http://data.cma.cn/). The dataset included minimum temperature (°C), maximum temperature (°C), average temperature (°C) and precipitation (mm).

According to phenological observations, the occurrence times of the same phenology event for 'SY63' exhibited inconsistency across site-years (Fig. 1d). Regreening took place between March 20 and August 20 (with a daylength of 12.0 h and 14.5 h), heading occurred between May 14 and September 30 (with a daylength of 11.6 h and 14.5 h), and physiological maturation varied from June 10 to November 18. Meanwhile, we observed significant temperature fluctuations throughout the entire developmental period. The difference in average temperature across site-years where heading occurred was approximately 3–8 °C (Fig. 1c).

2.2. Phenological models

Two kinds of models are often used for rice phenology predictions: temperature driven and temperature-photoperiod driven models (Caffarra et al., 2011; Cannell and Smith, 1983; Reaumur, 1735; Yin et al., 1997). Both kind of models use cardinal temperatures to calculate the daily thermal time, and use the accumulated thermal time to predict the occurrence of developmental stages. The difference is that temperature-photoperiod driven models adjust the daily thermal time during developmental stages that are sensitive to daylength, namely photoperiod effects. In addition, different functions have been developed to translate daily temperature to thermal time and daylength to photoperiod effects. The combinations of different functions resulted in different models. We tested 12 rice phenology models including three temperature driven models (Fig. 2, M1–3) and nine



(caption on next page)

Fig. 1. Geographical location of phenology observation sites (a), daylength hours by latitudes and dates (b), mean temperature (c) of the study area, and (d) phenological occurrence dates. (a) Different colors and symbol shapes indicate different numbers of observed years. (b) Daylength was calculated from the day of year and latitude. (c) Daily mean temperatures were calculated as averages across all sites during the growing seasons from 1984 to 2010. (d) The box charts showed the range of phenological occurrence dates, with black points indicating the outliers, red points indicating the means, and the black center lines indicating the medians. The left and right box boundaries are the 25th and 75th percentiles. The left and right whiskers represent the 5th and 95th percentiles.

temperature-photoperiod driven models (Fig. 2, M4–12). We divided the 12 models into three groups (Fig. 2, G1, G2, G3) based on different temperature response functions. Within the same model group, the temperature response function remained consistent, whereas the difference arose from the photoperiod response function.

2.2.1. *Temperature response functions*

The three types of models (M1, M2, M3) driven solely by temperature are shown in Fig. 2. For the two-segment piecewise linear temperature response function, as shown in Eq. (1) (Fig. 2, M1, M4, M5, M6), the development rate increases linearly from the base temperature (T_b) to the optimum temperature (T_o), and then remains constant at higher T.

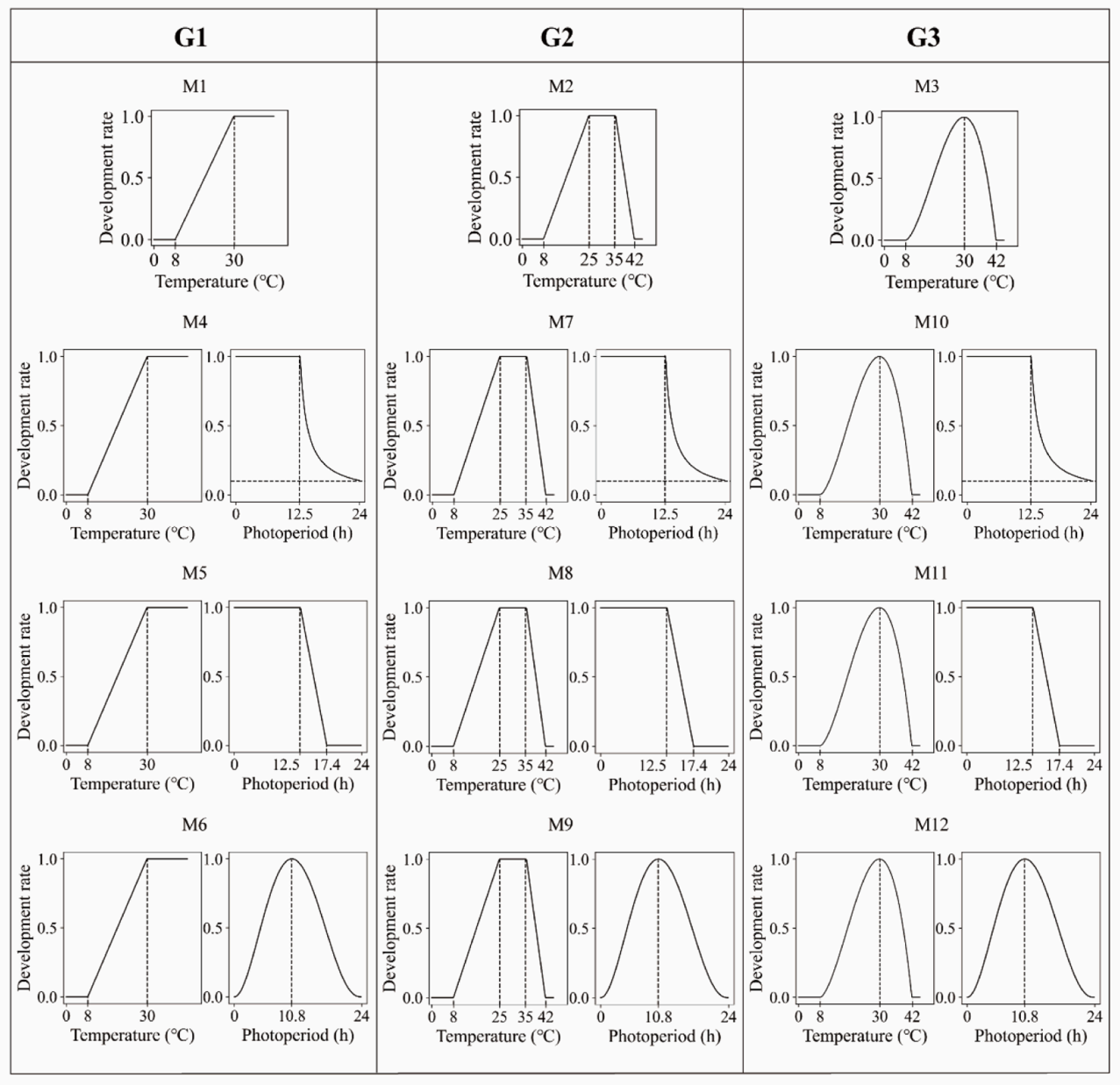


Fig. 2. An overview of 12 phenological models. M1–M3 represent temperature-driven phenological models, M4–M12 represent phenological models driven by both temperature and photoperiod. G1, G2, and G3 represent three distinct model groups. G1 used the two-segment piecewise linear temperature response function, G2 used the three-segment piecewise linear temperature response function, and G3 used the Beta temperature response function. The first model of each group disregarded photoperiodic effect, whereas the second to fourth models incorporated the two-segment piecewise nonlinear, the two-segment piecewise linear and the Beta photoperiod response function, respectively.

$$f_T(T,\ T_b,\ T_o) = \begin{cases} 0,\ if\ T\ <\ T_b \\ \dfrac{T-T_b}{T_o-T_b},\ if\ T_b\ \le\ T\ \le\ T_o \\ 1,\ if\ T\ >\ T_o \end{cases} \quad (1)$$

where f_T (0–1) represents the response of development rate to temperature, T is the daily average temperature.

For the three-segment piecewise linear temperature response function (Fig. 2, M2, M7, M8, M9), as shown in Eq. (2), the development rate increases linearly from T_b to the lower optimum temperature (T_l), and remains constant until the upper optimum temperature (T_h), then decreases linearly to 0 at the critical temperature (T_c).

$$f_T(T,\ T_b,\ T_l,\ T_h,\ T_c) = \begin{cases} \dfrac{T-T_b}{T_l-T_b},\ if\ T_b\ \le\ T\ \le\ T_l \\ 1,\ if\ T_l\ <\ T\ <\ T_h \\ \dfrac{T_c-T}{T_c-T_h},\ if\ T_h\ \le\ T\ \le T_c \\ 0,\ if\ T\ <\ T_b\ or\ T\ >\ T_c \end{cases} \quad (2)$$

For the Beta temperature response function, the effect of temperature on rice development rate is described by a Beta function (Wang and Engel, 1998). This function defines the response of development rate to temperature by three cardinal temperatures, T_b, T_o, and T_c (Eq. (3)).

$$f_T(T,\ T_b,\ T_o,\ T_c) = \begin{cases} \dfrac{2(T-T_b)^{\partial}(T_o-T_b)^{\partial}-(T-T_b)^{2\partial}}{(T_o-T_b)^{2\partial}},\ if\ T_b \le T \le T_c \\ 0,\ if\ T\ <T_b\ or\ T\ >\ T_c \end{cases}$$

$$\partial = ln2/ln[(T_c-T_b)/(T_o-T_b)] \quad (3)$$

where ∂ is a sensitivity parameter that describes the curvature of the temperature response function.

For the three temperature response functions mentioned above, the cardinal temperatures are T_b (8 °C), T_o (30 °C), T_l (25 °C), T_h (35 °C), and T_c (42 °C), as shown in Table 1.

2.1.2. Photoperiod response functions

Rice is widely recognized as a short-day crop, i.e., short daylength accelerates rice development, while long daylength slows development (Thomas and Vince-Prue, 1997; Yano et al., 2001). The photoperiod response only has developmental effects during a certain period that can vary across different models (Ahmad et al., 2012; Boogaard et al., 2014; Bouman et al., 2001; Gao et al., 1992; Meng et al., 2003). We assumed that the photosensitive period was from jointing to heading. Accordingly, we selected three widely used photoperiod response functions to quantify the effect of daylength on rice development rate (Boogaard et al., 2014; Bouman et al., 2001; Yin and Kropff, 1996).

For the two-segment piecewise nonlinear photoperiod response function (Fig. 2, M4, M7, M10), as shown in Eq. (4). Prior to the optimal day length (DL_o), the rate of development remains unaffected by the daylength (DL). Subsequently, when DL is longer than DL_o, the developmental rate exhibits a nonlinear decline, asymptotically approaching 0.1 (Jones et al., 2003; Sulṭānī and Sinclair, 2012; Zhang and Tao, 2013; Tong, 2020).

$$f(DL) = \begin{cases} 1,\ if\ DL\ \le\ DLo \\ \left[1+\dfrac{P2R}{136}\times(DL-DL_o)\right]^{-1},\ if\ DL\ >\ DL_o \end{cases} \quad (4)$$

where $f(DL)$ represents the effect of daylength (0–1). P2R is a light-sensitive parameter indicating the delay in panicle formation from ear

Table 1
Temperature-photoperiod function parameters.

Parameters	Description	Value	References
T_b	Lower threshold temperature for development	8 °C	van Oort et al. (2011)
T_o	Optimum temperature for development	30 °C	Bouman et al. (2001), van Oort et al. (2011), Boogaard et al. (2014), Wang et al. (2017)
T_l	Lower threshold of the optimum temperature range for development	25 °C	Jia (1989), Yin (1995), Bouman et al. (2001), Ahmad et al. (2012), Boogaard et al. (2014), Wang et al. (2017)
T_h	Upper threshold of the optimum temperature range for development	35 °C	Jia (1989), Yin (1995), Bouman et al. (2001), Ahmad et al. (2012), Boogaard et al. (2014), Wang et al. (2017)
T_c	Upper threshold temperature for development	42 °C	van Oort et al. (2011)
μ	Photoperiod sensitive coefficient	–15.46	Yin and Kropff (1996)
δ	Short photoperiod promotion coefficient	2.06	Yin and Kropff (1996)
ε	Long photoperiod suppression coefficient	2.48	Yin and Kropff (1996)
DL_c	Critical daylength for heading	16 h	Liang and Cai (1980)
DL_o	Optimum daylength for heading	12.5 h	Yin (1997), Jia (1989)

panicle formation.

For the two-segment piecewise linear photoperiod response function (Fig. 2, M5, M8, M11) that introduces a light-sensitive parameter (PPSE), as shown in Eq. (5), the development rate is not affected when $DL \le DL_o$, and then decreases linearly to 0 when $DL > DL_o$ as influenced by crop-specific PPSE and DL (Bouman et al., 2001).

$$f(DL) = \begin{cases} 1,\ if\ DL\ \le\ DL_o \\ 1-(DL-DL_o)\times PPSE,\ if\ DL\ >\ DL_o \end{cases} \quad (5)$$

PPSE was set as 0.2 through parameter optimization.

For the Beta photoperiod response function (Fig. 2, M6, M9, M12), the effect of daylength on rice development rate uses a Beta curve, as shown in Eq. (6). This relationship defines the minimum daylength and maximum daylength as 0 and 24 h, respectively (Yin and Kropff, 1996). The development rate first increases to the maximum, and then gradually decreases and tends to 0. Moreover, the development rate changes faster near the maximum, but changes slower at the edges of the function.

$$f(DL) = \exp(\mu)DL^{\delta}(24-DL)^{\varepsilon} \quad (6)$$

where μ, δ, and ε are model parameters.

2.3. Model calibration by thermal requirement

Model G1 and G3 calculate thermal requirement (°C d) from regreening, as shown in Eq. (7).

$$Thermal_D = \sum_{d=Reg}^{D}[f_d(T)\times\Delta T] \quad (7)$$

where D is subsequent phenological date after the regreening date. 'd=Reg' represents the regreening date, which is the starting date for calculations. ΔT represents the temperature difference. For Eq. (1), ΔT is calculated as the subtraction of T_b from T_o. For Eq. (3), ΔT is the temperature difference between T and T_b.

Model G2 calculate thermal requirement (°C d) from regreening, as shown in Eq. (8).

$$Thermal_D = \sum_{d}^{D} thermal_d$$

$$thermal_d = \begin{cases} 0, \ if \ T < T_b \ or \ T > T_c \\ T - T_b, \ if \ T_b \leq T \leq T_l \\ T_l - T_b, \ if \ T_l < T < T_h \\ (T_l - T_b) - \dfrac{T - T_h}{T_c - T_h} \times (T_l - T_b), \ if \ T_h \leq T \leq T_c \end{cases} \quad (8)$$

where $thermal_d$ is daily effective heat (°C d).

Model G1 and G3 calculate photo-thermal requirement (°C d) by multiplying daily thermal requirement by daily daylength effect $f_d(DL)$, as shown in Eq. (9).

$$Photothermal_D = \sum_{d=Reg}^{D} [f_d(T) \times \Delta T \times f_d(DL)] \quad (9)$$

Model G2 calculate photo-thermal requirement (°C d) as shown in Eq. (10).

$$Photothermal_D = \sum_{d=Reg}^{D} [thermal_d \times f_d(DL)] \quad (10)$$

We simulated rice phenology using three temperature-driven models (Fig. 2, M1, M2, M3) and nine temperature-photoperiod-driven models (Fig. 2, M4–M12) at three scales (the whole study region, the local-level, and the site-level). All simulations started from the observed dates of regreening to eliminate the impact of transplanting shock. The steps for calculating simulation errors for the whole study region were as follows (Fig. 3): Firstly, the entire growing season was divided into five development stages (D_1–D_5). Subsequently, TR (thermal requirement) for each development stages and model were calculated and calibrated. Then, we took the calibrated TR as the fixed parameter to simulate rice phenology. The crop advanced to the next phenological stage once the observed cumulative TR_{daily} (daily effective heat) surpassed the calibrated TR. The simulated date on which the crop completed a specific development stages was denoted as $Y_{i,sim}$, while the corresponding observed date was denoted as $X_{i,ob}$. The simulation error was the difference in days between the simulated and observed dates. Meanwhile, to ensure that none of the simulations failed to predict physiological maturation, we set the maximum duration from regreening to physiological maturation as 124 days, which was the longest duration in our records.

The methods for calibrating TR at both the region-level and the site-level were similar to above steps. The differences across the three geographic scales were as follows: the cluster number was 1 for the whole study region, the cluster number was 2–45 for the local-level, and the cluster number was 46 for the site-level. For the site-level calibration, there was no spatial variation, so we only focused on the inter-annual variation of TR. Site-specific parameters were tailored for each site. In contrast, for the entire study region and the local-level, we considered not only the inter-annual variation of TR, but also the spatial variation. The derived parameters were used for the clusters with analogous environmental conditions. The data analysis code in this study is available (https://github.com/Tanjiaojiao1/phenological_model).

2.4. Model performance evaluation

Simulation errors were calculated by subtracting observed dates from the simulated dates of the same development stage. Mean absolute error (MAE) was used to quantify the model performance, as shown in Eq. (9). The larger the value, the greater the error; the closer the value was to 0, the more accurate the simulation.

$$\mathrm{MAE} = \sum_{i=1}^{N} \left| Y_{i,sim} - X_{i,ob} \right| \Big/ N \quad (11)$$

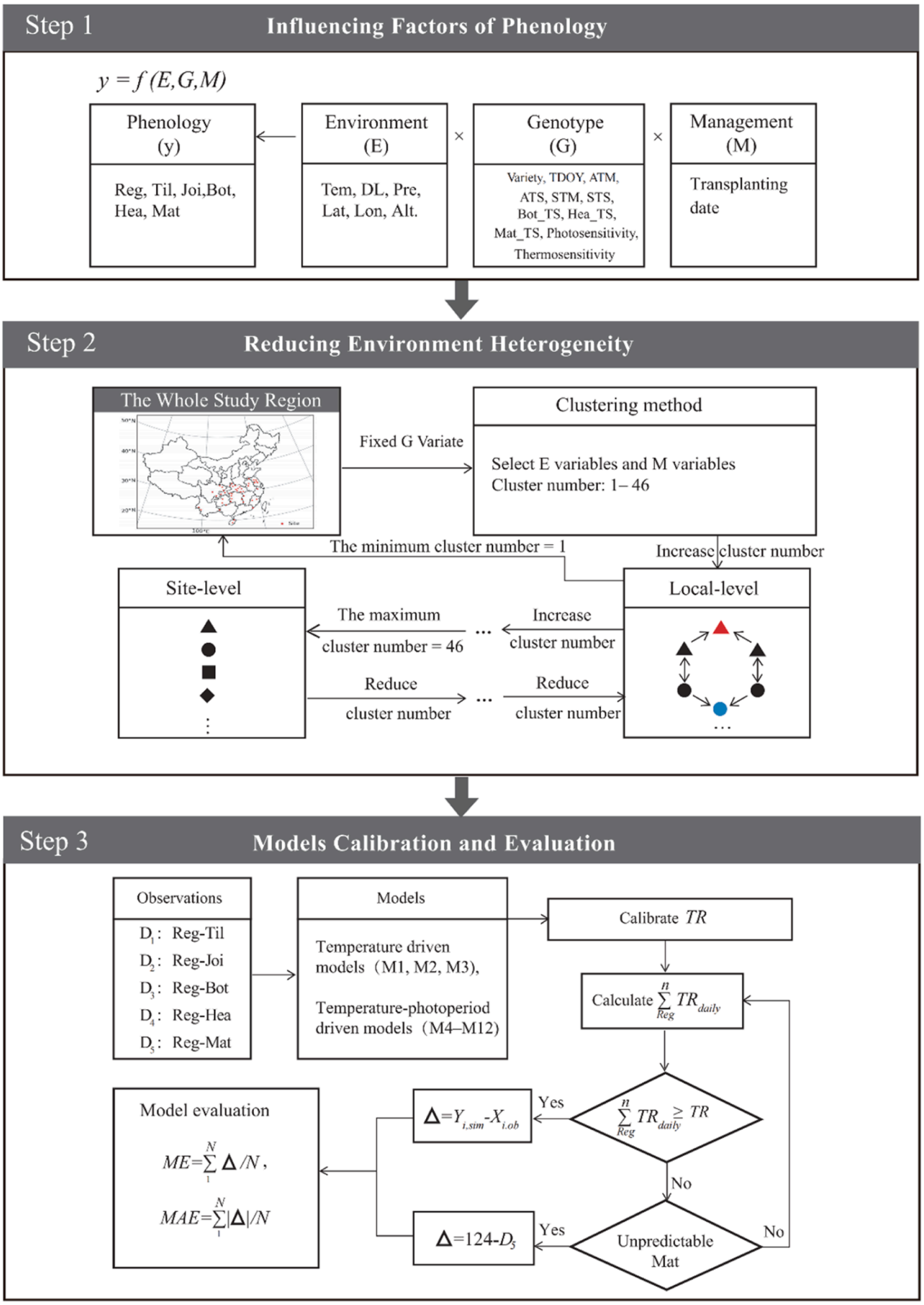
where $X_{i,ob}$ is the i th observed phenological date, $Y_{i,sim}$ is the i th simulated phenological date, and N is the number of observed phenological data.

2.5. Reducing simulation errors by clustering observation sites

Clustering analysis is an unsupervised learning algorithm that aims to classify data with similar characteristics into the same category. The object of clustering is to minimize intra-cluster sample differences while maximizing inter-cluster differences. K-means is one of the most commonly used clustering methods. This method assigns the samples into different categories based on the similarity of the distance between samples. Our objectives were to group sites with similar environments into the same cluster to reduce environmental heterogeneity, and to test whether clustering parameterization could reduce simulation errors. We used K-means clustering analysis from scikit-learn library in Python 3.11 based on climatic and geographical factors (Afrin et al., 2018).

The steps to optimize the clustering parameters were as follows. Firstly, we selected potential variables that influenced crop phenology based on expert knowledge and the characteristics of our observational data (Wallach et al., 2021a, 2021b, 2021c). These variables included geographic information (latitude, longitude, altitude), management practices (transplanting dates), and weather data (daily average temperature, thermal accumulation, daylength). Because our research crop was rice grown in paddy fields in southern China, and the variety was fixed as 'SY63', which excluded factors related to crop variety and water availability. Given the well-established notion that rice was widely known as a short-day crop and its development was influenced by photoperiod, we also considered daylength and its variation with different latitudes. Additionally, our data lacked records of management practices such as fertilization and pesticides, so we did not include these factors. The focus was on the impact of different environmental conditions on crop development. In the end, the set of potential variables included, latitude (Lat), longitude (Lon), altitude (Alt), 'Day of Year for Transplanting date' (TDOY), 'Annual Mean Temperature' (ATM), 'Annual Accumulation of Growing Degree Days' (ATS), 'Thermal Accumulation from Transplanting to Booting' (Bot_TS), 'Thermal Accumulation from Transplanting to Heading' (Hea_TS), and 'Thermal Accumulation from Transplanting to Physiological Maturation' (Mat_TS).

Secondly, we examined the correlation between potential variables and the days from transplanting date to physiological maturation date (Days_Trans_Mat). This analysis led to an exclusion of irrelevant variable Lon (r=−0.15, $P > 0.05$, Table S1). Subsequently, we quantified the impact of the remaining relevant independent variables on Days_Trans_Mat, establishing a hierarchy of importance: Mat_TM > ATM > Lat > Mat_TS > Alt > TDOY > ATS > Boot_TS > Head_TS (Table S2). However, considering that the values of Mat_TM, Mat_TS, Boot_TS and Head_TS relied on phenological observations, they did not be directly used as clustering parameters for phenological prediction. Therefore, we calculated 'Mean Temperature (STM)' and 'Thermal Accumulation' (STS) during the growing season (within 120 days after transplanting) independent of phenological observations, as substitutes for variables (such as, Mat_TS) that were dependently associated with phenological observations. Similarly, we used 'Thermal Accumulation within 60 days after transplanting' (TD60_TS) and 'Thermal Accumulation within 70 days after transplanting' (TD70_TS) instead of Bot_TS and Hea_TS, respectively (Jia, 1989; Wang, 2015). Subsequently, the inter-correlation among new variables was analyzed for selecting the



(caption on next page)

Fig. 3. An Overview of the proposed method to improve phenology models. Step 1: We first considered all variables that affected crop phenology, including environment, crop variety (genotype), and management practices. Step 2: The sites with homogeneous environmental conditions were divided into the same category by clustering to reduce environmental heterogeneity. The cluster number showed a gradual increase from the whole study region to the local-level to the site-level. ▲, ●, ■, ◆ represent individual sites with unique environmental conditions. ▲ and ● represent different clusters divided by the clustering method. ↔ signifies the process of distinguishing sites with environmental heterogeneity. Step 3: D_1–D_5 represent five developmental periods. TR represents the calibrated thermal requirement. TR_{daily} represents daily effective heat. The simulated date on which the crop completed a specific stage was denoted as $Y_{i,sim}$, while the corresponding observed date was denoted as $X_{i,ob}$. Δ is the difference in days between simulated and observed dates.

appropriate clustering variables (Fig. 4). It was noted that the selected variables should be correlated to accurately group sites with similar environments into the same category. However, highly correlated variables may mistakenly cluster similar sites into different categories. Therefore, the variables used for clustering should have weak correlations. Based on Fig. 4, we preliminarily identified that STM, Lat, Alt, TDOY, or the combinations of these variables could be used as clustering variables.

Thirdly, we quantified the spatial variability of correlated variables to estimate whether there was spatial heterogeneity (as shown in Table S3). Because our observations were based on the site-level, there was inherent spatial heterogeneity. Root mean squared error and variation coefficient were used to directly quantify the variability of latitude (Lat) and altitude (Alt), while the multi-year average values of RMSE and variation coefficient were calculated for STM and TDOY.

Finally, we classified 46 sites in southern China using STM, Lat, Alt, TDOY, and the combinations of these variables, respectively. We used Inertia value to determine an approximate range of cluster number, which provided a broader scope for comparing Mean Absolute Error of different clustering paramaters. We focused on the MAE of the physiological maturation stage and throughout all developmental stages (encompassing tillering, jointing, booting, heading, and physiological maturation). This was because that physiological maturation stage played a decisive role in the final yield, and the average predictive error of all developmental stages was also one of the key indicators to evaluate model performance. According to the principle of minimum MAE and convenient application, we determined the optimal clustering variables and delineated the most appropriate range of cluster number within our clustering analysis.

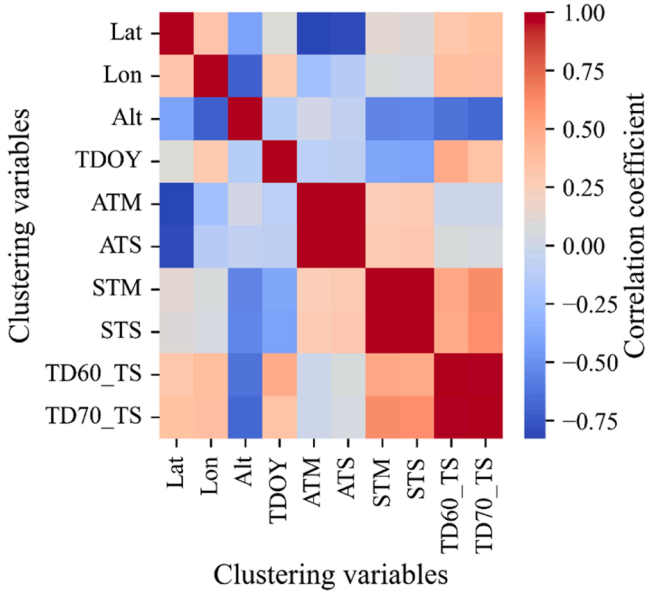


Fig. 4. Correlation coefficient matrix of the important environmental variables. The shade of the cell color corresponds to the strength of the correlation, with darker shade indicating stronger correlations. Variables include Latitude (Lat), Longitude (Lon), Altitude (Alt), The Day of Year for Transplanting date (TDOY), Annual Mean Temperature (ATM), Annual Accumulation of Growing Degree Days (ATS), Mean Temperature (STM) and Thermal Accumulation (STS) during the growing season (within 120 days after transplanting), Thermal Accumulation within 60 days after transplanting (TD60_TS), Thermal Accumulation within 70 days after transplanting (TD70_TS).

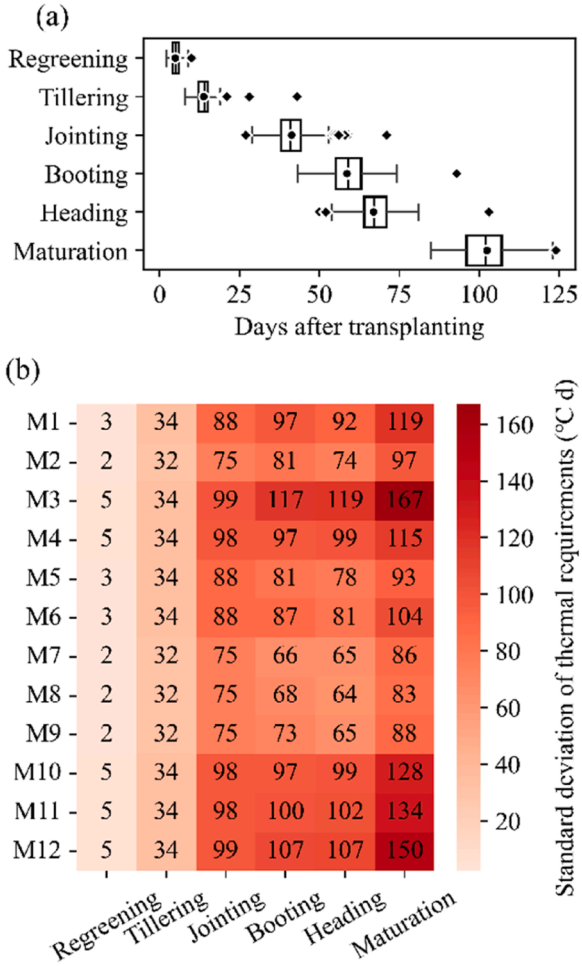


Fig. 5. The observation of rice phenological development stage and the standard deviation of thermal requirements (TR_{std}) across site-years. Panel (a) showed the durations from transplanting date to the dates of subsequent phenological events across site-years, and panel (b) showed the standard deviation of thermal requirements (TR_{std}) across site-years for all models. In panel (a), boxes and whiskers represent the 25th–to–75th and the 10th–to–90th percentile ranges, respectively, of the duration from transplanting date to subsequent phenological dates. The thick lines and black dots in the boxes are the median and mean of duration, respectively. The black diamonds outside the boxes are the duration values beyond the 10th–to–90th percentile range. In panel (b), M1–M12 represent the 12 phenology models, with detailed descriptions given in Fig. 2 and Table 1. The values in the cells of panel (b) correspond to the TR_{std}. The cell colors represent the value of TR_{std} across site-years, where darker colors indicate higher variability.

3. Results

3.1. Variability of thermal requirements across observation sites

Fig. 5 illustrates the significant variability observed in the occurrence time of different stages and thermal time requirements (TR) across site-years. As rice development progressed, differences in the time of occurrence increased incrementally (Fig. 5a). For instance, the days from transplanting to regreening ranged from 3 to 11 days, with an average of 6 days. The duration from transplanting to heading varied between 51 and 82 days, with an average of 68 days. The period from transplanting to physiological maturation ranged from 86 to 124 days, with an average of 103 days.

We used TR_{std} to quantify the variability in thermal requirements across site-years for each model. The temperature variations resulted in site-specific patterns of daily thermal accumulation for TR. As rice development progressed, the standard deviation of thermal requirements (TR_{std}) across site-years for all models became more pronounced with crop development (Fig. 5b). Taking M3 as an example, the TR_{std} across site-years was 34 °C d at tillering, whereas TR_{std} increased to 167 °C d at physiological maturation. Among the 12 models, M3, M10, M11, and M12 exhibited larger TR_{std}. These four models were based on the Beta temperature response function (Fig. 2, G3). In contrast, models employing the three-segment piecewise linear temperature response function (Fig. 2, G2), such as M2, M7, M8, and M9, showed smaller TR_{std}. Furthermore, photoperiod can affect the accumulation of daily thermal units. Models incorporating a photoperiod effect (M5–M12) exhibited lower TR and smaller TR_{std} compared with models without a photoperiod effect (M1–M3). For example, at physiological maturation, the TR of M4–M12 ranged from 1396 °C d to 1759 °C d, whereas the TR of M1–M3 ranged from 1656 °C d to 1844 °C d (Fig. S1). The TR_{std} across site-years for M10–M12 ranged from 128 °C d to 150 °C d, whereas the TR_{std} across site-years for M3 was 167 °C d.

3.2. Optimal clustering parameters in our study area

The inertia values indicate that the optimal cluster number varies across different variables (Fig. 6). Mean Absolute Errors (MAE) across a range of 1 to 24 clusters were used to measure the performance of different variables, including 'STM', 'Lat', 'Lat+STM', 'Lat+STM+Alt', 'Lat+STM+TDOY', and 'Lat+STM+TDOY+Alt'. The appropriate ranges for clusters number were found to be 6–12, 8–15, 12–18, 18–24, 18–24, and 20–30, respectively. Fig. 7 showed the changes in MAE with different clustering variables at different development stages and the overall mean. For example, at the physiological maturation stage, using 'STM' as the clustering variable with 9 clusters resulted in a minimum MAE of 4.6 days (Fig. 7a). To determine the best cluster numbers and variables, the variations in MAE for all stages (tillering, jointing, booting, heading, and physiological maturation) were shown in Fig. 8. Although 'STM' achieved quick convergence with an MAE of 4.0 days when the cluster number was 6, its MAE did not decrease with an increasing cluster number. However, when cluster number exceeded 6, the MAE values of 'Lat+STM', 'Lat+STM+Alt', and 'Lat+STM+TDOY+Alt' were lower than that of 'STM'. When the cluster number was 12, the MAE values of 'Lat+STM', 'Lat+STM+Alt' and 'Lat+STM+TDOY+Alt' were 0.4 days lower than that of 'STM'. Considering the variables 'Lat+STM', 'Lat+STM+Alt' and 'Lat+STM+TDOY+Alt', they performed similarly in terms of predictive precision. However, 'Lat+STM+TDOY+Alt' included two additional variables, 'Lat+STM+Alt' included one additional variable, compared to 'Lat+STM'. Therefore, this work selected 'Lat+STM' as the optimal variable with a cluster number of 12 considering the trade-off of accuracy and efficiency.

3.3. Improving model performance by reducing environmental heterogeneity

The performance of different models across the whole study region

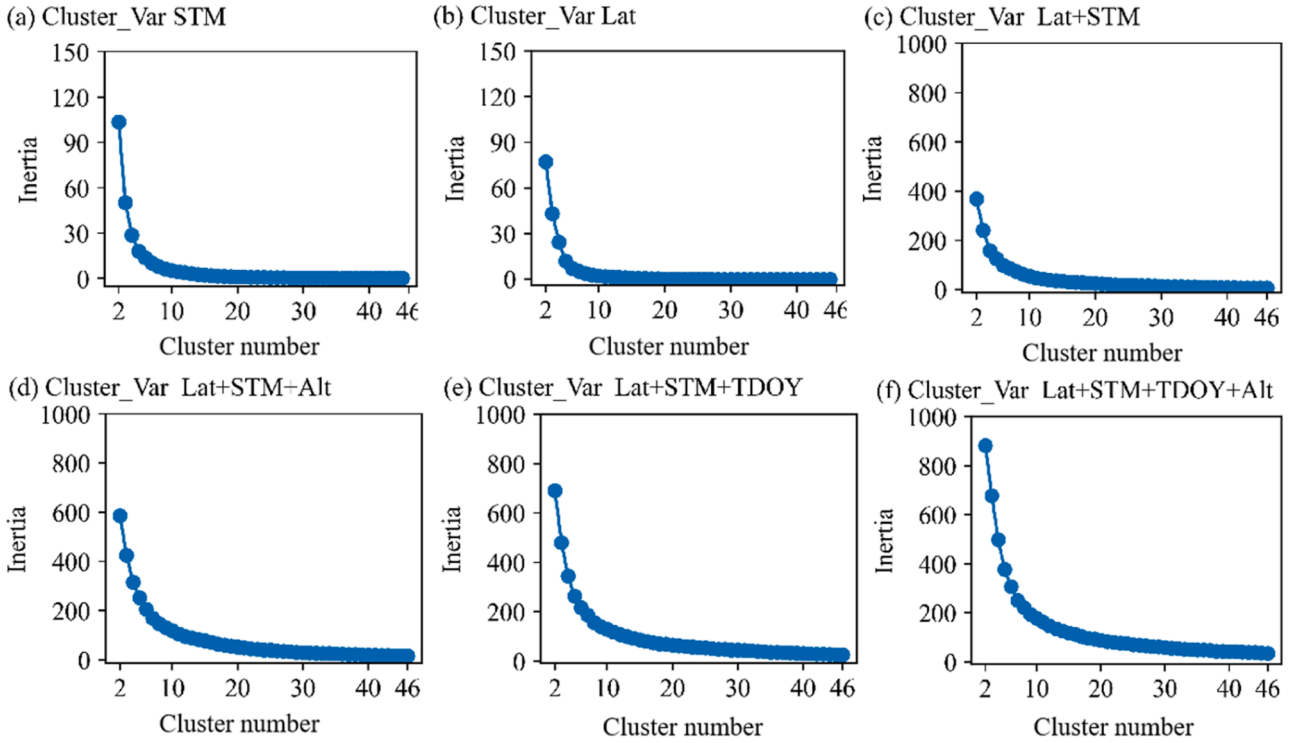


Fig. 6. Response of Inertia value to cluster number and variable combination. STM represents the mean temperature within 120 days after transplanting. Lat represents latitude. Alt represents altitude. TDOY represents the day of year for transplanting date. Inertia indicates the effects of clustering on variable variability. The lower Inertia values indicate smaller variability within clusters.

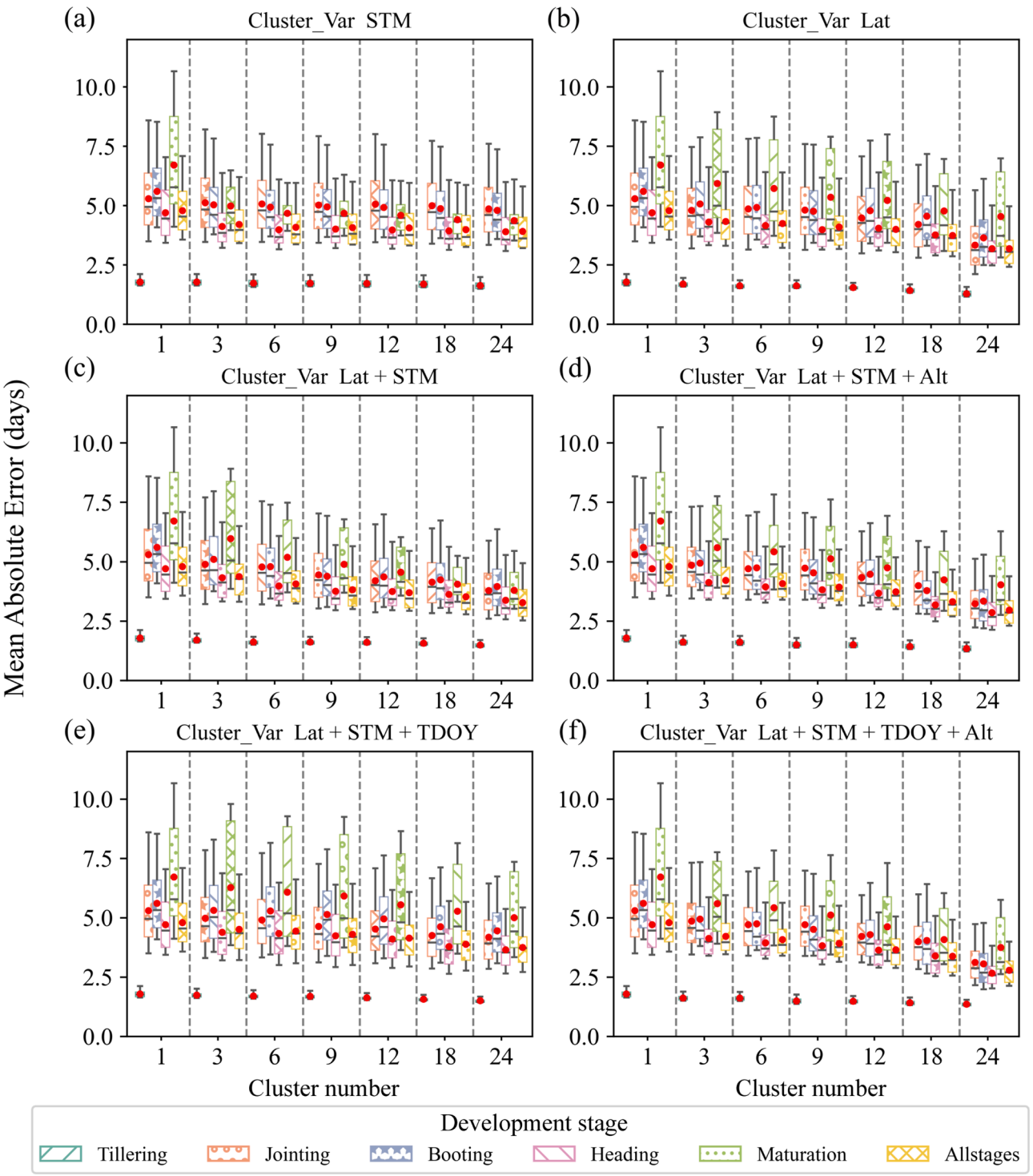


Fig. 7. The change mean absolute error (MAE, days) with different clustering variables at different development stage and all stage mean. 'Allstages' represents mean absolute error of all stages (tillering, jointing, booting, heading, and physiological maturation).

was shown in Fig. 9. The MAE of Models across site-years ranged from 1.6 to 10.5 days with decreasing performance with the development of rice. Specifically, the ranges of MAE for models at different development stages were as follows: 1.6–2.0 days (at tillering), 3.5–8.6 days (at jointing), 4.1–8.5 days (at booting), 3.4–7.1 days (at heading), and 4.0–10.5 days (at physiological maturation). The ensemble model produced MAE values of 1.8, 5.3, 5.6, 4.7, and 6.6 days for each development stage, respectively. Among the models, M3 and M10–M12 showed the highest MAE, while M2 and M7–M9 performed better. However, the best and worst models differed for different development stages. For example, for jointing and booting, M2 had the smallest MAE (3.5 and 4.1 days). For heading, both M8 and M9 had the smallest MAE (both 3.4 days). For physiological maturation, M8 had the smallest MAE (4.0 days). Across all development stages (allstages), M2 and M9 consistently produced the smallest MAE of 3.6 days.

Increasing cluster number significantly reduced the MAE for all models and all development stages (Fig. 7). Based on 'Lat+STM' clustering analysis (Fig. 7c), MAE for tillering, joining, heading, and

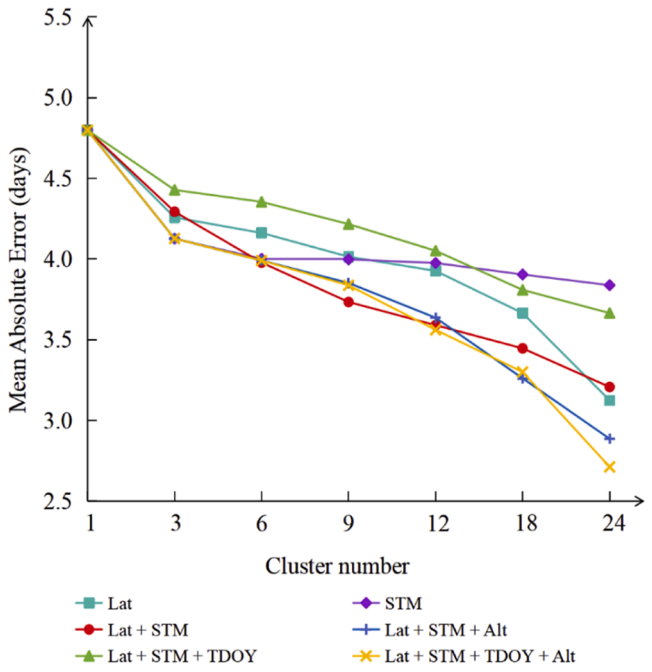


Fig. 8. The decreasing trend of mean absolute error (MAE, days) for different clustering variables across all stages.

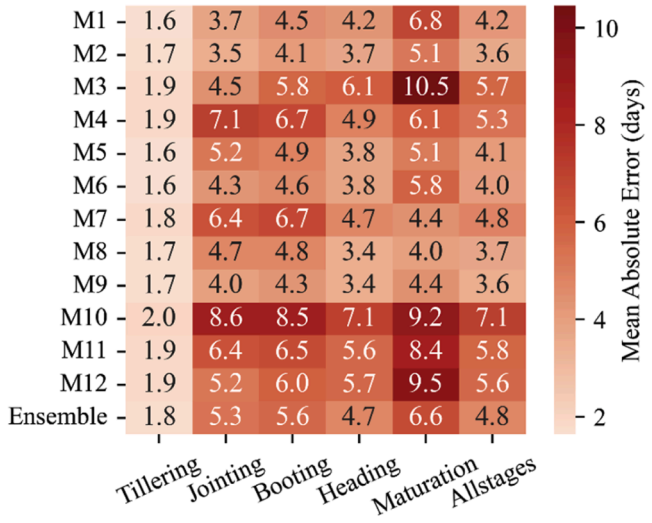


Fig. 9. Mean Absolute Errors (MAE, days) for all development stages and models across the whole study region. M1–M12 represent the 12 phenology models, with detailed descriptions given in Fig. 2. The last row (Ensemble) represents the MAE of all models (M1–M12), and the last column (Allstages) represents the MAE across all development stages (including tillering, jointing, booting, heading, and physiological maturation). The numbers within each cell correspond to MAE values. Cells with a darker shade indicate higher Mean Absolute Error (MAE) values.

physiological maturation stages exhibited a consistent decrease as cluster number increased from 1 to 24. Specifically, tillering stage was the first phenological stage in our simulation, and its MAE displayed a minimal variation across cluster number 1–24, with only a 0.4-day difference. For the jointing stage, the MAE decreased from 5.3 days in the initial cluster number 1 to 3.7 days in the final cluster number 24, a reduction of 1.6 days. Similar patterns were observed for the booting, heading, and physiological maturation stages, with total decreases of 1.7, 1.4, and 2.9 days, respectively. The MAE of all stages decreased significantly from 4.8 days in cluster number 1 to 3.2 days in cluster number 24. The physiological maturation stage showed the most significant improvement as increasing number of clustering categories. As cluster number continuously increased, the MAE of physiological maturation stage progressively converged towards the MAE of all stages. Within an increment in cluster number from 1 to 24 categories, the rate of MAE reduction for all development stages decelerated progressively.

3.4. Performance of different model groups across two scales

Model G1 (two-segment) and G2 (three-segment) demonstrated similar performance levels in both site-level and region-wide simulations, whereas Model G3 (Beta) exhibited the poorest performance (Fig. 10). As the study area expanded from the site-level to the whole study region, the performance gap between high-performing and low-performing model groups became more pronounced (Fig. 10). For the site-level simulations, the Mean Absolute Error (MAE) at physiological maturation stage was 1.7 days (ranging from 0 to 10 days) for G1, 1.8 days (ranging from 0 to 10 days) for G2, and 3.4 days (ranging from 0 to 23 days) for G3. For the whole region-wide simulations, the MAE at physiological maturation stage was 6.0 days (±5.6 days) for G1, 4.5 days (±4.1 days) for G2, and 9.4 days (±8.0 days) for G3. Model G3 consistently displayed higher MAE values across all scales. Notably, models with larger MAE, such as G3, experienced greater reductions in MAE after clustering compared to models with initially lower MAE. For instance, MAE of M2 was reduced by 3.3 days (from 5.1 to 1.8 days) at physiological maturation stage when moving from the whole study region to the site-level, whereas MAE for Model M3 was reduced by 7 days (from 10.5 to 3.5 days).

When comparing the effect of with and without the photoperiodic effect for models, models that included a photoperiod function exhibited higher accuracy than those without a photoperiod function. For example, for the whole study region, MAE for M2, M7, M8 and M9 was 5.1, 4.4, 4.0 and 4.4 days in G2, respectively. Similarly, MAE for M3, M10, M11 and M12 was 10.5, 9.2, 8.4 and 9.5 days in G3, respectively (Fig. 10b). Moreover, photoperiod functions had varying effects on model performance at different scales. At the site-level, the model performance mainly depended on the structure of the temperature response function, with the photoperiod having minimal impact (Fig. 10a). However, as the simulation expanded from site level to regional scale, the positive effect of photoperiod on model accuracy gradually became apparent. Taking the simulation errors of physiological maturation in G1 as an example, we compared M1, which did not consider the photoperiodic effect, with M4, which incorporated it. At the site level, the MAE of M1 at physiological maturation was 1.8 days, while M4 also had an MAE of 1.8 days. However, for the whole study region, M1 had an MAE of 6.8 days, compared to M4, which had an MAE of 6.1 days.

4. Discussion

4.1. Site-specific thermal requirements reveal the limitations of phenological models

We adequately described TR for different development stages for rice variety 'SY63' at 46 sites during 1984–2010 by using 12 phenological models. We observed variations in TR across site-years (Fig. 5b, Fig. S1). Our results are consistent with those of previous studies that have also reported site-specific TR (Fadón et al., 2023; Tao et al., 2012; van Oort et al., 2011; Zhang et al., 2008). This variability arises from crops adjusting their phenological events in response to changes in temperature and daylength within their specific growing environments.

Specifically, phenology is influenced by various environmental factors such as temperature fluctuations, seasonal changes, and photoperiod (Cleland et al., 2007; Lieth, 1974; Siebert and Ewert, 2012; Wang et al., 2017). To mitigate the impact of environmental changes on crop development, different sites adopt appropriate management measures,

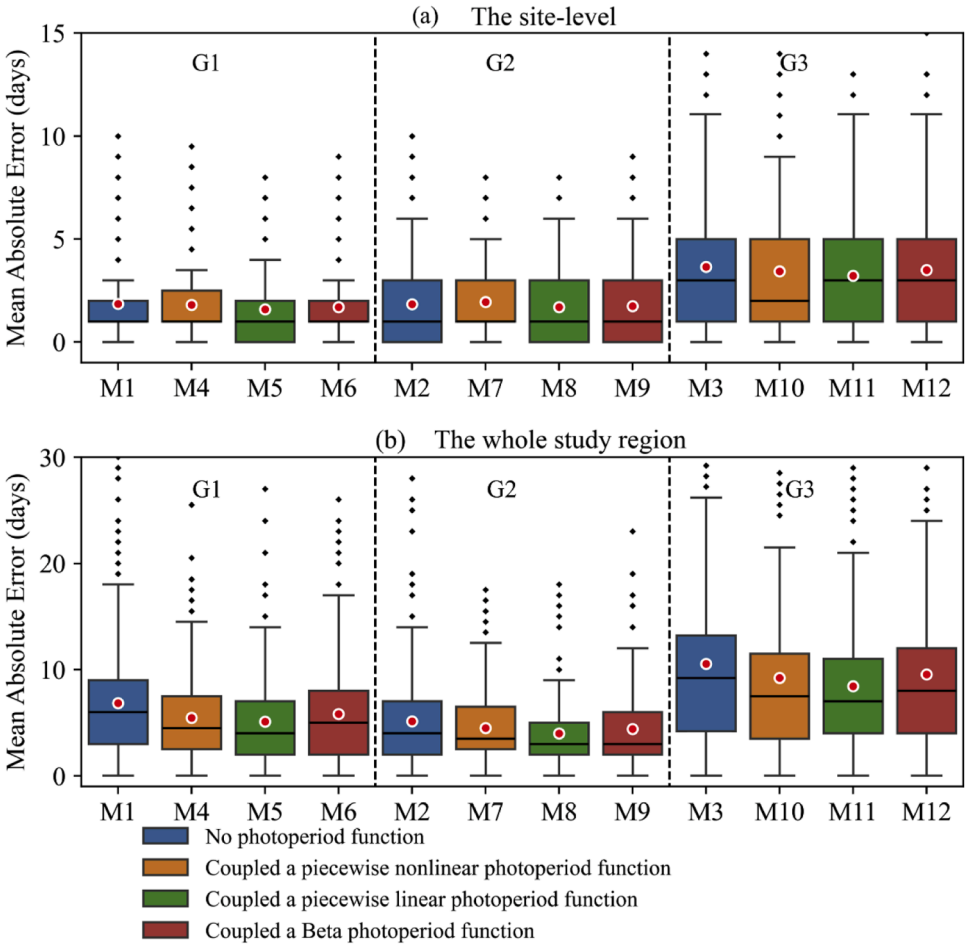


Fig. 10. Comparison of mean absolute error (MAE, days) for models at physiological maturation stage. M1–M12 represent the 12 phenological models, with detailed descriptions given in Fig. 2. G1, G2, and G3 represent three model groups. (a) MAE for the site-level. (b) MAE for the whole study region. Boxes of the same color represent models that use the same photoperiod response function but different temperature response function. Boxes and whiskers represent the 25th–to–75th and the 10th–to–90th percentile ranges of MAE, respectively. The thick lines in the boxes represent the median values. The red dots represent the mean values. The black diamonds represent MAE values beyond the 10th–to–90th percentile range.

such as adjusting planting dates (Wood et al., 2014). We observed that planting dates ranged from late-March to mid-July across site-years, and the number of days after the same phenological event after transplanting differed by approximately 10–40 days across site-years (Fig. 5a). This result can be attributed to the wide range of latitudes and topographic fluctuations in Southern China, resulting in significantly different mean growing season temperatures for 'SY63' across site-years. For instance, sites where heading occurred around August 1 experienced temperatures that were 5–8 °C warmer than observed at other sites where heading occurred at different times (Fig. 1c, d). Furthermore, photoperiod varied from 11.6 to 14.5 h over the study area (Fig. 1b), while standard deviations of the thermal requirements (TR_{std}) at heading stage varied between 64 °C d and 119 °C d (Fig. 5b). Additionally, latitude has been shown to impact TR. Liu et al. (2013) demonstrated that a 1° increase in northern latitude resulted in a decrease of 25.6 °C d in TR. Therefore, the contribution of daily TR to the development rate varies under different environmental conditions. These results suggest that TR is affected not only by the genetics of crop varieties, but also by site-specific factors.

The variation in TR acrosssite-years challenges the widely accepted concept that TR is solely determined by variety-specific traits and genetics (Hill and Li, 2022; Wang, 1960). This discrepancy may arise from the limitations of models that fail to account for crop adaptation to their specific cultivated environment. It has been reported that plants exhibited environmental plasticity in various phenotypic traits (Dingkuhn et al., 2005; Pigliucci, 2005). For instance, the distribution of roots within the soil profile can vary to accommodate natural fluctuations in water and nutrient availability (Hinsinger et al., 2009). In response to changes in ambient temperature, the temperature response of photosynthesis was observed to undergo modifications over different time scales (Kumarathunge et al., 2019). Wu et al. (2019a) reported that incorporating a plastic temperature response function into the temperature sensitivity of growth rate reduced simulation errors by half at physiological maturation. These results highlight the crucial need for future research to determine model enhancements that will consider crop adaptation mechanisms to diverse environments.

However, previous studies that demonstrated the environmental variability of crop model parameterization often relied on a single model or only considered a limited number of models to calculate TR (Campoy et al., 2012; Fadón et al., 2023, 2020; Gannouni et al., 2017; Viti et al., 2010). As a result, the uncertainties associated with these models have not been thoroughly investigated or addressed. In order to comprehensively assess the accuracy and consistency of phenological models and to investigate the sources of model uncertainties, we selected 12 different phenological models, including three temperature-driven models and nine temperature-photoperiod-driven models. We divided the 12 models into three groups (G1–G3, Fig. 2) based on the different methods used to model crop development rates in response to temperatures. The distinctions between different model groups were attributed to their respective temperature response functions. Compared with models

based on temperature alone, models that considered both photoperiod and temperature improved the accuracy of phenological predictions at scales larger than sites. On the other hand, models with the piecewise linear temperature response function performed better than those with the Beta function (Figs. 2 and 10, G1 and G2 vs. G3). This result can be attributed to the different mechanisms used in these models to describe the response of crop development rate to temperature. The Beta temperature response function used a fixed optimum development temperature (30 °C), whereas three-segment piecewise linear function had a range of 25–35 °C for optimum development temperature. When the optimum temperature point was exceeded, the development rate of the Beta function dropped significantly. Poor-performing models tended to underestimate crop development rates under non-optimal temperatures (Wang et al., 2015), resulting in predicted durations longer than the actual observed durations. These findings align with previous research that highlighted the potential limitations in accurately describing crop development rates under non-optimal temperatures (Tao et al., 2012; Wang et al., 2017; Zhang and Tao, 2013).

The differences within the same model group could be attributed to the photoperiod response functions. Specifically, the first model of each model group disregarded photoperiodic effect, whereas the second to fourth models incorporated the two-segment piecewise nonlinear, the two-segment piecewise linear and the Beta photoperiod response functions, respectively. For the entire growing season, models employing the two-segment piecewise linear photoperiod response function (M5, M8, and M11) exhibited smaller error (Fig. 10). This is because the Beta photoperiod response function exhibits the least sensitivity to daylength when compared with the other photoperiod response functions, and its accuracy depends on the temperature response function. This is consistent with the findings of Wang et al. (2015). The two-segment piecewise nonlinear photoperiod response function is very sensitive to daylength changes near DL_o, after which the rate slows down, resulting in significant differences in photo-thermal requirement changes. Moreover, optimizing parameters using default thresholds derived from limited environments also contributed to increased simulation errors. Previous studies have indicated that phenological models calibrated only within specific temperature ranges might become less accurate with climate warming. Iizumi et al. (2009) reported regional divergences in the posterior distribution of parameters, such as the base temperature and the curvature factor of spikelet sterility caused by low temperature. Similarly, van Oort et al. (2011) also demonstrated that models calibrated using default parameter thresholds may not accurately simulate the duration from emergence to flowering in a warming climate.

Although there are differences among phenological models, as the scale extends from the site level to the whole study region, the errors of these models show a consistent and gradually increasing trend. Interestingly, we also observe consistency in model performance at different scales. Models utilizing the three-segment piecewise linear (G2 in Figs. 2 and 10) temperature response function are consistently found to have the best performance. In contrast, models utilizing the Beta temperature response function (G3 in Figs. 2 and 10) are consistently considered to be the least accurate. This result contrasts with previous research results that emphasized the importance of selecting specific models based on the local crop planting system in a particular region (Zhang and Tao, 2013). The main reason for this difference is that we do not differentiate early rice, late rice, and single-season rice in our study, as our data primarily focuses on single-season rice. Additionally, our study area is the subtropical monsoon climate region of southern China, excluding the temperate monsoon climate region of northeastern China. Previous studies have typically calibrated all parameters based on site-level or local-level data (Wang et al., 2015; Zhang and Tao, 2013). In contrast, we simulate rice phenology based on variety-specific thermal requirements at specific scales. Our method has advantages in scale specificity, variety adaptability, and simplified parameter calibration, and enables more accurate and adaptable simulation of rice phenology across different varieties and environmental conditions.

4.2. *Clustering method reduces environmental heterogeneity and improves phenological model performance accuracy*

Phenological models have most often been developed and validated under stable and homogenous field conditions (Manivasagam and Rozenstein, 2020; Pasquel et al., 2022). However, as these models are increasingly applied at larger scales beyond individual fields, their accuracies can be affected by the failure to account for crop adaptation mechanisms to different environments. Consequently, phenological simulation errors can vary among different models and study regions. When applying to a broader scale, yet these issues have been received limited attention. In this study, we focus on increasing the accuracy of phenology models across various spatial scales by using a clustering method to identify and group homogeneous sites. We aim to address key aspects such as determining optimal clustering variables and the appropriate cluster number to increase simulation accuracy.

The clustering method can mitigate uncertainties arising from variations in topography, geographical location, and environmental heterogeneity in models (Hazeu et al., 2010). By carefully selecting appropriate clustering variables and categories, diverse environmental factors can be integrated into process-based models as optimized parameters. Clustering homogeneous sites into the same category where a crop exhibits similar adaptability to its environment effectively compensates for the absence of crop adaptation processes in phenology models, thereby reducing simulation errors. This method can improve the performance of phenology models in regions characterized by heterogeneity in environmental conditions. Our findings demonstrate that using Lat and STM as clustering variables significantly reduce simulation errors. STM is the most important variable affecting development rate. And Lat can indirectly affect phenology by temperature and photoperiod. For example, Liu et al. (2013) investigated the phenological response of maize to environmental changes across different latitudes in China. They observed an increase in the length of the vegetative growth period and a decrease in the length of the reproductive growth period with higher northern latitudes. Similarly, according to the findings of the China Hybrid Rice Meteorological Research Cooperation Group (1981), during the period from April to October (when there was minimal temperature variation between the northern and southern regions), an increase of 1° in north latitude resulted in a 19 °C d increase in the TR for 'SY63' from planting to heading, accompanied by a 0.11-hour extension in photoperiod. Moreover, it should be noted that environmental conditions and crop adaptability vary with regions. Therefore, when applying this clustering method to other regions, it is crucial to readjust and optimize the clustering parameters based on the specific characteristics of the study area. Specifically, the cluster number needs to be reparametrized, and the selection of clustering variables should consider region-specific factors such as altitude, soil moisture, soil characteristics, water resource availability, and other relevant factors.

While the clustering method effectively reduces errors in phenological simulations caused by environmental heterogeneity, it does not account for time heterogeneity, thereby failing to adequately address the errors arising from heterogeneity. Various time-related factors, including seasonal fluctuations and climate changes can significantly impact crop growth and development. For example, temperature fluctuations, daylength, and soil nitrogen availability throughout seasons can impact crop development rates and the timing of phenological events (Liang and Cai, 1980; Ritchie and Nesmith, 1991; Roberts and Summerfield, 1987; Wong and Nortcliff, 1995). Moreover, climate change introduces variability associated with environmental conditions, such as temperature, humidity, and precipitation, further affecting crop growth and development (Shu et al., 2022; Smith et al., 2013; Tao et al., 2008). Additionally, other time-related factors, such as alterations in agricultural management practices and varieties, can reduce the accuracy of phenological simulations (Wood et al., 2014; Zhao et al., 2022).

These factors exhibit variations associated with different times and locations, resulting in spatio-temporal disparities in the occurrence of phenological events. Further work should be done by combining time-dependent data analysis methods, such as time series analysis, dynamic modeling, and hybrid models, to better interpret and predict phenological variability.

4.3. The effect of climate warming on phenology may be overestimated

Most regions around the world have experienced a warming trend during the past few decades, and this phenomenon is expected to accelerate in the future (Calvo Buendía et al., 2020; Hansen et al., 2006; Huang et al., 2017; Shu et al., 2022). Therefore, understanding how crop phenology responds to and adapts to climate warming is a prerequisite for accurately assessing crop yields in the future. In theory, phenological models predict crop phenology based on certain TR values, and the time required to accumulate TR will decrease with warming, resulting in a shorter duration of crop development days. Previous studies have indicated that under future climate warming scenarios, process-based crop models generally simulate advancement in phenological events, shortened developmental period, and decreased yields (Cleland et al., 2007; Li et al., 2015; Tao et al., 2008; Wang et al., 2015). However, some studies have also indicated that the shortened developmental period and decreased yields due to climate warming were effectively mitigated by shifting crop varieties and management practices (Porker et al., 2020; Seyoum et al., 2018; Zhao et al., 2023, 2022). However, previous studies do not disentangle the effects of crop variety shifts from multiple factors, nor do they consider crop adaptability to climate change. Therefore, the modeling assumption of constant thermal requirement leads to theoretical predictions that do not fully align with the actual situations.

Our study results confirm that the thermal requirement of a specific rice variety 'SY63' is significantly site-specific. Phenology simulation errors increase with the expansion of the study area as the result of crop models failing to reflect the mechanisms of crop adaptation to different growing environments. Wu et al. (2019b) also showed that increased temperatures did not universally result in advanced phenological dates and shortened developmental periods, demonstrating the adaptability of crops to climate warming, even when varieties were not changed. Additionally, differences in the temperature responses of different models at non-optimum temperatures become more pronounced as climate warms (Li et al., 2015). Wang et al. (2017) also confirmed that at non-optimum temperatures, phenology simulated by different models deviated from each other and did not match observations. These results indicate that model parameters based on existing experiments will no longer be applicable as climate warms. Temperature is one of the main drivers of phenology, but other environmental factors, such as high-temperature stress and extreme precipitation, will also have a great impact on crop growth and development as the climate warms (Fu et al., 2023; Hossain and Teixeira Da Silva, 2012; Ihsan et al., 2016; Sharma et al., 2016). Currently, phenological models cannot adequately quantify the comprehensive effects of various environmental factors on phenology. Thus, the imperfect phenological models tend to overestimate the effects of climate warming on crop growth and development.

4.4. Limitations

The accuracy of models depends on the optimization of the input data and parameters. Optimizing model parameters by using many observed phenology datasets can produce output parameters that accurately reflect real planting conditions, thereby enhancing a model's accuracy and reliability. One of the limitations of our study is the observational bias in the phenological dataset. Phenological observations are collected on entire rice fields where phenological dates are determined and recorded when 50 % of the plants have reached a specific phenological stage (China Meteorological Administration, 1993). However, rice plants in the field do not enter the same phenological stage simultaneously, and the observed data cannot precisely indicate the specific times of developmental stages. For example, it typically takes about 30 days from the first effective tillering to reach the maximum tiller count. If we assume that simulations start from the observation date of the first effective tillering, and there will be deviations in the actual durations from tillering to subsequent phenological stages.

Management practices like transplanting date have a major impact on rice development, and we have tried to use it as the starting point of the initiating simulations. Due to varying degrees of human-induced damage observed after transplanting of rice, the period for growth recovery ranged from 3 to 11 days (Fig. 5a). To mitigate errors associated with the transplanting shock and its impact on simulation accuracy, we selected the regreening date as the reference point for initiating phenological simulation. And the change in regreening dates also served as an indicator of adjustment made to transplanting dates. Moreover, we considered the effect of the transplanting date when selecting clustering variables, as shown in Fig. 4. The adjustment of the transplanting dates was contingent upon geographical and climatic factors (such as, latitude, growing season average temperature, precipitation and sunshine hours), which collectively affected phenology. However, we found that incorporating transplanting dates as clustering variables did not enhance the simulation performance across 12 phenological models (Fig. 7). For the physiological maturation stage, the MAE of 'Lat+STM' showed a marked improvement, decreasing from 5.9 days in cluster number 3 to 3.7 days in cluster number 24, reducing the error by 2.2 days. Similarly, the MAE of 'Lat+STM+TDOY' variables showed a reduction from 6.2 days in cluster number 3 to 4.9 days in cluster number 24, indicating a 1.3-day decrease in error for physiological maturation stage. Hence, we decided that Lat and STM were more critical than transplanting for clustering.

We recognized that nitrogen limitation impacted crop growth and development, potentially resulting in delayed phenological stages. However, the study area experienced excessive nitrogen due to prolonged and heavy fertilization practices, leading to significant N_2O emissions and nitrogen-related environmental loading (Peng et al., 2006; Shang et al., 2014; Shen and Zhang, 2006). Consequently, despite variations in nitrogen management practices, we postulated that rice cultivation in this region was not nitrogen-constrained and excluded this factor from our modeling framework. Although other management measures (apart from transplanting date adjustment, variety alternation, and nitrogen fertilization) can affect model simulations, extensive studies have shown that their influence on crop phenology is minor (Zhao et al., 2013; Piao et al., 2019; Mahmood et al., 2022). Crop development is predominantly limited by temperature, day length, and nitrogen supply, and this limitation does not detract from the study's objective of optimizing phenological models.

Considering the environmental heterogeneity across various regions, it is crucial to recognize that the clustering parameters optimized for our specific study area are not inherently suited for wider applications. The diversity in climatic conditions, soil types, geography, and human management practices across different areas can significantly influence the performance of these parameters. When applying the clustering method to other regions, we need to re-optimize the region-specific clustering parameters (as shown in Fig. S2).

5. Conclusion

The significant variation in thermal requirements observed for a specific rice variety across different sites indicates the strong influence of environmental conditions on crop development. This result reveals the limitations of phenological models based on the constant thermal requirement theory that fail to effectively capture crop responses and adaptations to environmental changes. Accordingly, we propose a clustering method aimed at reducing environmental heterogeneity. By

incorporating latitude and growing season mean temperature as key clustering variables and selecting an appropriate cluster number, we effectively improve the accuracy of phenological prediction in a region of diverse environmental conditions. However, the simulation performances of 12 models are different, and these differences primarily arise from their different temperature response functions. Additionally, the effect of photoperiod on simulation accuracy is more evident at scales larger than site level. When applying models to large geographic areas encompassing diverse climatic zones, the uncertainty of phenological simulations will increase due to diverse environmental conditions. Further work is required to validate the clustering method with more varieties in more regions, and to improve models by coupling a machine learning approach to effectively reflect crop adaptation mechanisms to environmental changes.

CRediT authorship contribution statement

Jiaojiao Tan: Writing – review & editing, Writing – original draft. **Gang Zhao:** Writing – review & editing, Methodology, Funding acquisition. **Qi Tian:** Writing – review & editing. **Lei Zheng:** Writing – review & editing. **Xiaofeng Kang:** Conceptualization. **Qinsi He:** Writing – review & editing. **Yu Shi:** Writing – review & editing. **Bin Chen:** Writing – review & editing. **Dingrong Wu:** Writing – review & editing, Data curation. **Ning Yao:** Writing – review & editing. **Liang He:** Conceptualization, Writing – review & editing. **Amit Kumar Srivastava:** Writing – review & editing. **Yi Li:** Conceptualization. **Jianqiang He:** Conceptualization. **Hao Feng:** Conceptualization. **Qiang Yu:** Supervision, Software, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

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